

Panagiotis Lymperopoulos

PhD Candidate in Computer Science, Expected Graduation May 2026.
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Education

Tufts University

August 2019 - June 2026

PhD Candidate in Computer Science
MS in Data Science

University of California, Los Angeles (UCLA)

September 2015 - June 2019

Bachelor of Science in Computational and Systems Biology

Relevant Coursework: Deep Learning, Machine Learning for Graph Data, Reinforcement Learning, Natural Language Processing. **Skills:** Python, PyTorch, TensorFlow, Reinforcement Learning, Deep Learning, Probabilistic Modeling, Graph Neural Networks, RAG, LLMs, LLM Agents, Agile Development.

Work Experience

Research Intern, Skyfall, New York City, US (Remote)

June 2025 - August 2025

- Developed World Models for LLM-based planning agents by integrating foundation models' implicit world knowledge with external and domain-specific data.
- Combined RAG, LLM-as-judge and probabilistic inference on graphical models to automatically design and optimize sparse world models for planning with MCTS.

LLM Engineer, Agora Labs, Rome, Italy (Remote)

June 2024 - May 2025

- Led team of developers in development and deployment of customer-facing text-to-SQL and automated data preprocessing application.
- Designed and implemented information extraction pipeline for medical data, integrating traditional NLP workflows (NER, entity-linking) with LLMs.
- Focused on ensuring data privacy and security by deploying and fine-tuning local, open-source LLMs.

Data Science Intern, Optechain, Athens, Greece

June 2021 - August 2021

- Implemented ML pipeline for Data Cleaning, Training and Deployment of time-series prediction and computer-vision models on AzureML
- Developed infrastructure on the entire stack for querying deployed models, submitting results to database and serving them to end-users.

Research Highlights

Act-to-Ground: Neurosymbolic Grounding for Planners

September 2023 - Present

- Developed Act-to-Ground (A2G), a framework for training neural grounding models for symbolic planners using weak supervision from environment interaction or demonstrations. Motivated by a need to separate perception from action to ensure safe and interpretable decision making.
- Proposed a novel MCMC sampler and a score-matching objective to improve the stability and performance of neurosymbolic grounding models.
- Presented the framework and its successful evaluation in planning domains at the 19th Conference on Neurosymbolic Learning and Reasoning (NeSy 2025).

Uncertainty Quantification for Tool-Using LLMs

April 2024 - October 2024

- Developed a novel method for quantifying the uncertainty of Large Language Model (LLM) question-answering systems that use external tools such as RAG or querying ML models.
- Motivated from increasing need of safety in deployments of LLMs in critical domains (e.g. medicine).
- Presented results at the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025).

Open-world Agent Modeling

May 2023 - September 2023

- Developed a novel method for online modeling of agent behavior from observations, that accounts for unobservable agent-internal states, crucial in HCI, HRI applications.
- Presented a paper describing the method and its results at the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2024).

Accelerating Infeasibility Analysis with Deep Learning

September 2022 - September 2023

- Created a graph neural network-based algorithm for accelerating infeasibility analysis of boolean constraint satisfaction problems.
- Achieved more than two orders of magnitude improvement on MUS enumeration for hard problems and presented findings at AISTATS 2024.

Publications

- Lymperopoulos, P., Qiu, H., & Min, B. (2020). Concept wikification for COVID-19. *Proceedings of the 1st Workshop on NLP for COVID-19 (Part 2) at EMNLP 2020*. <https://www.aclweb.org/anthology/2020.nlpCOVID19-2.29>
- Brummer, A. B., Lymperopoulos, P., Shen, J., Tekin, E., Bentley, L. P., Buzzard, V., Gray, A., Oliveras, I., Enquist, B. J., & Savage, V. M. (2021). Branching principles of animal and plant networks identified by combining extensive data, machine learning and modelling. *Journal of the Royal Society Interface*, 18(174), 20200624.
- Gopalakrishnan, S., Soni, U., Thai, T., Lymperopoulos, P., Scheutz, M., & Kambhampati, S. (2021). Integrating planning, execution and monitoring in the presence of open world novelties: Case study of an open world monopoly solver. *CoRR*, abs/2107.04303. <https://arxiv.org/abs/2107.04303>
- Lee, A. H., Lymperopoulos, P., Cohen, J. T., Wong, J. B., & Hughes, M. C. (2021). Forecasting covid-19 counts at a single hospital: A hierarchical bayesian approach. *Poster in ICLR 2021 Workshop on Machine Learning for Preventing and Combating Pandemics*, arXiv preprint arXiv:2104.09327.
- Polleys, C. M., Lymperopoulos, P., Thieu, H.-T., Genega, E., Liu, L., & Georgakoudi, I. (2021). Deep-learning-based image restoration of depth-resolved, label-free, two-photon images for the quantitative morphological and functional characterization of human cervical tissues. *Imaging, Manipulation, and Analysis of Biomolecules, Cells, and Tissues XIX*, 11647, 116470Z.
- Feeney, P., Schneider, S., Lymperopoulos, P., Liu, L., Scheutz, M., & Hughes, M. C. (2022). Novelcraft: A dataset for novelty detection and discovery in open worlds. *arXiv preprint arXiv:2206.11736*.
- Panagiotis Lymperopoulos, L. L., Yukun Li. (2022). Exploiting variable correlation with masked modeling for anomaly detection in time series. *NeurIPS 2022 Workshop on Robustness in Sequence Modeling*. <https://openreview.net/forum?id=TCJuZs585W>
- Goel, S., Lymperopoulos, P., Thielstrom, R., Krause, E., Feeney, P., Lorang, P., Schneider, S., Wei, Y., Kildebeck, E., Goss, S., et al. (2024). A neurosymbolic cognitive architecture framework for handling novelties in open worlds. *Artificial Intelligence*, 104111.
- Lymperopoulos, P., & Liu, L. (2024). Graph pruning for enumeration of minimal unsatisfiable subsets. *International Conference on Artificial Intelligence and Statistics*, 2647–2655.
- Lymperopoulos, P., & Sarathy, V. (2025). Tools in the loop: Quantifying uncertainty of llm question answering systems that use tools. *arXiv preprint arXiv:2505.16113*.